Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Leila Hernandez	lherna@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

help students stay organized and manage their time more effectively by creating a Study Planner website.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Several existing websites and apps help students organize their assignments, manage their time, and keep track of deadlines

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

Study Planner designed to help students organize their schoolwork and manage their time more effectively. Users will be able to add assignments, enter due dates, organize tasks by subject, set study goals, and mark assignments as completed. The website will also display upcoming and overdue assignments so students can prioritize their work. I want students to use this application as a central place to track their academic responsibilities and reduce

stress.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The goal of my program is to take user input about assignments, subjects, due dates, and study goals, process and store the information using dataclasses, lists, loops, and conditional statements, and display organized assignments, overdue tasks, and progress summaries to help students manage their schoolwork efficiently.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

assignment titles, subjects, due dates, and completion status. They can also enter study goals and target study hours.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will store assignments and study goals in lists of dataclass instances, use loops to display and organize assignments, and use conditional statements to identify completed or overdue assignments. It will also count completed assignments and compare due dates to help students prioritize their work and track their progress.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

Users will receive an organized dashboard displaying their assignments, subjects, due dates, overdue tasks, and completion status. They will also see their study goals and a progress summary showing how many assignments they have completed

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Assignment, Study Goal, Subject, and Student. The Assignment dataclass will store the assignment title, subject, due date, and completion status. The Goal dataclass will store the study goal, target study hours, and completion status. The Subject and Student will help organize assignments and personalize the study planner.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need:
- Assignment title
- Subject or class name
- Due date
- Completion status (completed or not completed)
Details I will ignore:
- Teacher's name
- Assignment location or classroom
- Grade received
- Textbook or materials required

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The most difficult parts of this project will be connecting the different website routes and managing the assignment and study goal lists using dataclasses. Also making sure the program updates the state correctly when users add assignments or mark them as completed.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

Will use LLM to help me understand Python and Drafter concepts, troubleshoot errors, and plan my website.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

Building something I can use for next semester

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

want to build a Study Planner website to help students organize their assignments, track due dates, set study goals, and monitor their progress. What are three common problems students face when managing their schoolwork, and what are three simple features my website could include to help solve these problems?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The AI suggested that common problems students face include forgetting assignment deadlines, struggling to manage their time, and having difficulty tracking their academic progress. It recommended adding features such as an assignment tracker with due dates, a study goals section, and a progress dashboard showing completed and overdue assignments. Existing solutions mentioned included MyStudyLife, myHomework Student

Planner, Google Calendar, and Todoist. I plan to use these ideas to create a simple, user-friendly Study Planner website that helps students stay organized and balance their schoolwork.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I think creating a detailed progress dashboard with automatic tracking and advanced statistics might be too complicated for an introductory programming project because it could require complex calculations and additional features. I want to keep my website simple and focus on the Python and Drafter concepts I have learned in class. Could you suggest a simpler alternative?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

my project idea has become more focused on creating a simple Study Planner website that helps students manage their assignments and deadlines. Instead of building an advanced progress dashboard with complicated statistics, I will focus on basic features such as adding assignments, organizing them by subject, tracking due dates

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I may use ChatGPT to help me understand errors and TA office hours.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

first read the error message carefully to identify the problem. Next, I will check my code for syntax errors, incorrect variable names, missing type annotations, or problems with my dataclasses and routes. I will test the program in small sections and use unit tests to check that my functions work correctly. If the problem continues, I will consult my class notes and documentation, ask my instructor or TA for help, or use ChatGPT to explain the error without

blindly copying a solution. If I have a deployment problem, I will check that my files are uploaded correctly, my main.py file runs, and my GitHub repository is configured properly. Finally, I will test the website again locally and after deployment to make sure everything works as expected.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.